Building Circuits – Week 8

Record all measurements made as part of the lab under the relevant section. Graphs of current or voltage versus resistance when relevant are encouraged.

Set up your breadboard the same way you did last week: connect one of the 5V pins on the RPi to the + column and one of the ground pins to the – column. Run connectors from the + and – columns to rows on the main part of the breadboard.

1. LED in a circuit

An LED (light emitting diode) is a component that only allows current to flow in one direction, and it emits light when current passes through it. Like all diodes, it has a positive side (called the anode, the longer wire) and a negative side (called the cathode, the shorter wire).

a. Build a circuit with a resistor of at least 150Ω and an LED in series, connected between the +5V and ground rows.

b. How does the diode need to be oriented? Which wire on the LED goes to the +5V side and which goes to the GND connector?

   The negative side needs to be oriented towards the +5V side of the circuit, the positive wire goes towards ground.

c. What is the voltage drop across the resistor? Was this what you expected?

   Voltage drop of around 1.4-1.5 volts across the resistor. We expected that the drop would be 5V to satisfy Kirchhoff's Voltage Law.

d. What is the voltage drop across the LED?

   3.5V to satisfy Kirchhoff's Voltage Law.

e. Try removing the resistor from the circuit, keeping the circuit closed – the LED is just in series with the 5V supply.

f. What do you think will happen to the LED brightness?

   Without a resistor it will be able to pull significantly more current from the supply, resulting in a much brighter bulb

   NOTE: Do not leave the LED in this circuit closed for very long.

g. What important purpose does the resistor serve?

   The resistor provides resistance so that the amount of current running through the circuit does not convert into thermal energy causing issues.

h. Try including resistors of different values – how does LED brightness change vs resistor strength?

   Higher value resistor results in a dimmer light as the current is reduced

i. Measure the voltage drop across the resistor and LED in each case and make note of voltage drop versus resistance.

   100ohm: -0.7 (Resistor), -4.3 (LED)

   1kohm: -2.2(Resistor), -2.8 (LED)

j. Using the configuration with the highest LED brightness, move the 5V connection on the RPi to one of the 3.3V pins. What do you expect to happen to the LED brightness?

   We expect the brightness to go down as less potential energy can be converted into light

k. Move the RPi voltage connection back to 5V.

Does as expected 🙂 (Makes the led brighter when moving back to 5v)

2. Boost converter (step-up voltage)

A boost converter is a small circuit that takes a lower input voltage and converts it to a higher output voltage. You will use one to take the 5V from the RPi and increase it. This is the same kind of component we will use in a future lab to provide the higher voltage needed to power a radiation detector.

l. Add your step-up circuit component to increase your RPi voltage from 5V. Do not close your circuit yet.

m. Use the voltmeter to examine the output voltage of the step-up component and dial it up or down until the output is above 5V but below 10V.

n. Using the dimmest LED configuration from before (meaning select the appropriate resistor from those you tried previously), connect the LED circuit to the boost converter output instead of the RPi 5V directly.

o. How does the LED brightness change as you increase the output voltage?

   As the output voltage increases, so does the led brightness, this makes sense due to ohms law. As V increases and R stays the same, I will increase, resulting in more charge going through the LED allowing for more light to be produced.

p. How does the voltage drop across the resistor and LED change?

   Kirchhoff's law states that the voltage drop across both must equal the total voltage, as voltage increases, the drop on each must increase proportionally.

q. Do any of these results change with different color LEDs? Specifically, do any voltage drop values change? Is the relative brightness similar for different color LEDs?

   Some colors may appear dimmer due to their varying wavelengths storing more potential energy.

   NOTE: The boost converter works by rapidly switching current on and off internally. This creates a steady higher voltage on the output, but it also produces small, fast fluctuations called "ripple" or "noise" riding on top of that steady voltage. For powering an LED this does not matter, but in a future lab when we use a boost converter to power a sensitive detector, this noise will become very important. Keep this in the back of your mind.

3. Photo-diode

A photo-diode is a type of diode that generates a small electrical current when light hits it. Unlike the LED (which produces light from current), the photo-diode does the opposite – it converts light into current. Photo-diodes are used in "reverse bias" mode, meaning you connect the voltage backwards compared to how you connected the LED. In reverse bias, very little current flows normally, but when light hits the diode, it creates a small current that is proportional to the amount of light.

a. Replace the LED with a photo-diode. Start with the step-up component outputting 5V.

   NOTE: Photo-diodes operate in reverse bias mode so you will need to orient the diode the opposite way from how you had the LED.

b. What is the voltage across the resistor when you have a 5V bias and the photo-diode is exposed to the ambient light in the room?

   In ambient light with 5V bias, the voltage reads to be 3mV

c. What happens if you cover the photo-diode with your hand? What happens if you increase the bias voltage to 10V?

   The voltage decreases when covered. When the bias voltage doubles, the voltage through the resistor nearly doubles as well.

d. What is the "dark current" for this photo-diode? Dark current is the tiny amount of current that flows through the diode even when no light is hitting it. Use the voltage across the resistor and Ohm's law to calculate the current flowing through the diode.

   V=IR 0.00003 mV=I*100 0.0000003A

e. Is 5V enough supply voltage to see a signal from this diode?

   No

f. Try shining the flashlight from your phone onto the diode. What is the current output? Again, use the voltage across the resistor to determine this.

   0.020mV=100*I

   0.020/100=I

   0.0002A

g. Are you able to determine the saturation current for the photo-diode? Saturation current is the maximum current the diode will produce no matter how much more light you add.

   No, because we can not optimize the photon density going into the diode. Also likely depends on the output of the light itself.

4. Looking ahead

The photo-diode you just worked with produces a current when it detects light. Next week, you will work with a similar kind of sensor – a PIN diode – that is sensitive enough to detect individual flashes of light produced by radiation interacting with a crystal. The key challenge you will face is that the signal from a bare diode like this is extremely small. You will use an amplifier circuit to make that signal large enough to see on an oscilloscope. You will also use a boost converter like the one from today's lab to power a radiation detector, and the filtering concepts from last week's simulation will become directly relevant for getting a clean signal.

Sounds good